Network Layer

IPv4 ADDRESSES:

- An IPv4 address is 32 bits long.
- The IPv4 addresses are unique and universal.
- If a protocol uses N bits to define an address, the address space is 2^N because each bit can have two different values (0 or 1) and N bits can have 2^N values.
- The address space of IPv4 is 232 or 4,294,967,296.

Binary Notation:

- So it is common to hear an IPv4 address referred to as a 32-bit address or a 4-byte address. The following is an example of an IPv4 address in binary notation:

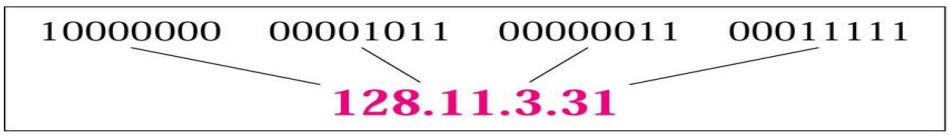
01110101 10010101 00011101 00000010

Dotted-Decimal Notation

- Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes. The following is the **dotted~decimal** notation of the above address:

117.149.29.2

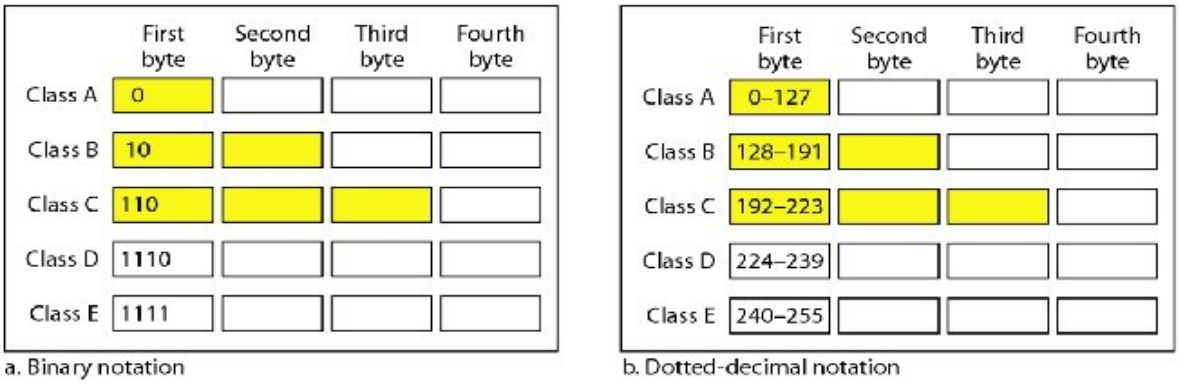
Dotted-decimal notation and binary notation for an IPv4 address



Classful Addressing

- **In classful addressing, the address space is divided into five classes: A, B, C, D, and E.**

Finding the classes in binary and dotted-decimal notation



Find the class of each address.

a. 00000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111

Solution

a. The first bit is O. This is a class A address.

b. The first 2 bits are 1; the third bit is O. This is a class C address.

c. The first byte is 14 (between 0 and 127); the class is A.

d. The first byte is 252 (between 240 and 255); the class is E.

- **In c1assfnl addressing, a large part of the available addresses were wasted.**

Netid and Hostid

Class	*Binary*	*Dotted-Decimal*	*CIDR*
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Default masks for classful addressing

- In classful addressing, an IP address in class A, B, or C is divided into netid and hostid.
- **In class A, one byte defines the netid and three bytes define the hostid.**
- **In class B, two bytes define the netid and two bytes define the hostid.**
- **In class C, three bytes define the netid and one byte defines the hostid.**

Mask

- Length of the netid and hostid (in bits) is predetermined in classful addressing, we can also use a mask (also called the default mask).
- A 32-bit number made of contiguous 1's followed by contiguous 0's.
- The mask can help us to find the netid and the hostid.
- For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the netid; the next 24 bits define the hostid.
- The last column of Table shows the mask in the form */n* where *n* can be 8, 16, or 24 in classful addressing.
- This notation is also called slash notation or Classless Interdomain Routing (CIDR) notation.

Subnetting

- If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets) or, in rare cases, share part of the addresses with neighbours.
- **Subnetting increases the number of 1's in the mask.**

Supernetting

- The time came when most of the class A and class B addresses were depleted; however, there was still a huge demand for midsize blocks.
- The size of a class C block with a maximum number of 256 addresses did not satisfy the needs of most organizations.
- Even a midsize organization needed more addresses. One solution was supernetting.
- In supernetting, an organization can combine several class C blocks to create a larger range of addresses.
- In other words, several networks are combined to create a supernetwork or a supemet. An organization can apply for a set of class C blocks instead of just one. For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks.
- The organization can then use these addresses to create one supernetwork.
- **Supernetting decreases the number of 1's in the mask.**

- For example, if an organization is given four class C addresses, the mask changes from /24 to /22.

Address Depletion

- The flaws in classful addressing scheme combined with the fast growth of the Internet led to the near depletion of the available addresses.
- **Classful addressing, which is almost obsolete, is replaced with classless addressing.**

Classless Addressing

- To overcome address depletion and give more organizations access to the Internet, classless addressing was designed and implemented.
-

DELIVERY

- The network layer supervises the handling of the packets by the underlying physical networks.
- We define this handling as **the delivery of a packet**.

Direct Versus Indirect Delivery

Direct Delivery

- In a direct delivery, the final destination of the packet is a host connected to the same physical network as the deliverer.
- Direct delivery occurs when the source and destination of the packet are located on the same physical network or when the delivery is between the last router and the destination host.

Indirect Delivery

- If the destination host is not on the same network as the deliverer, the packet is delivered indirectly.
- In an indirect delivery, the packet goes from router to router until it reaches the one connected to the same physical network as its final destination.

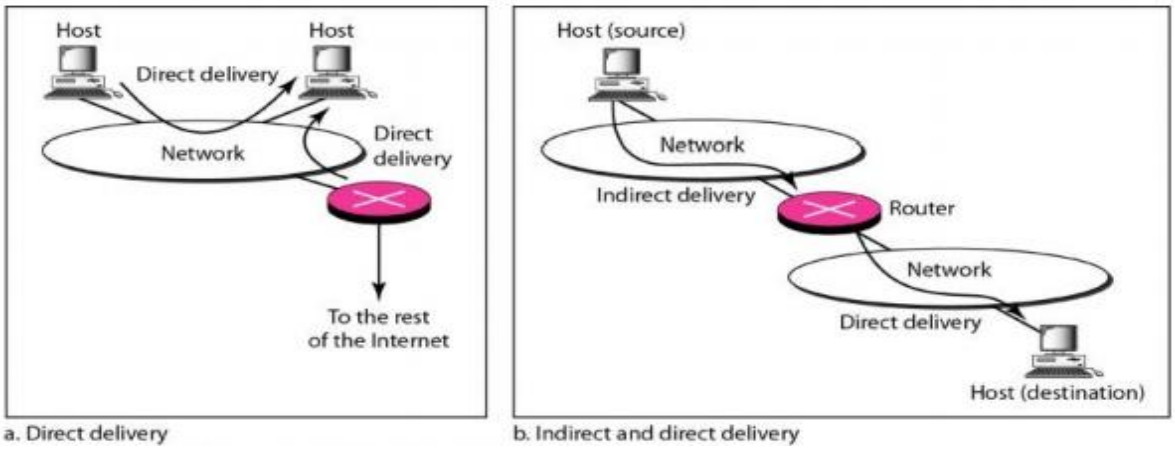


Figure: Direct and indirect delivery

FORWARDING

- Forwarding means to place the packet in its route to its destination.
- Forwarding requires a host or a router to have a routing table.
- When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this table to find the route to the final destination

Forwarding Techniques

Next-Hop Method Versus Route Method

- One technique to reduce the contents of a routing table is called the next-hop method.
- In this technique, the routing table holds only the address of the next hop instead of information about the complete route.

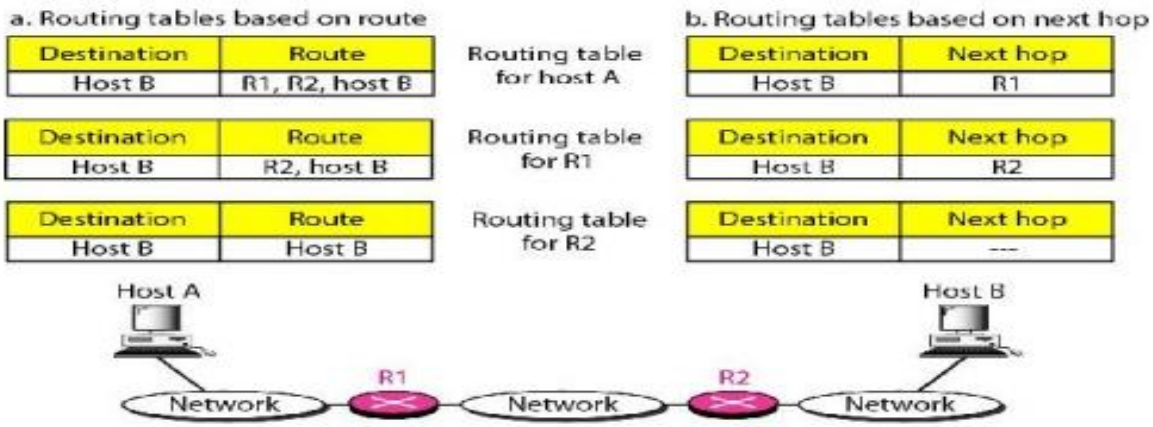


Fig: Route method versus next-hop method

Network-Specific Method Versus Host-Specific Method

- A second technique to reduce the routing table and simplify the searching process is called the network-specific method.
- Instead of having an entry for every destination host connected to the same physical network (host-specific method), we have only one entry that defines the address of the destination network itself.
- In other words, we treat all hosts connected to the same network as one single entity.

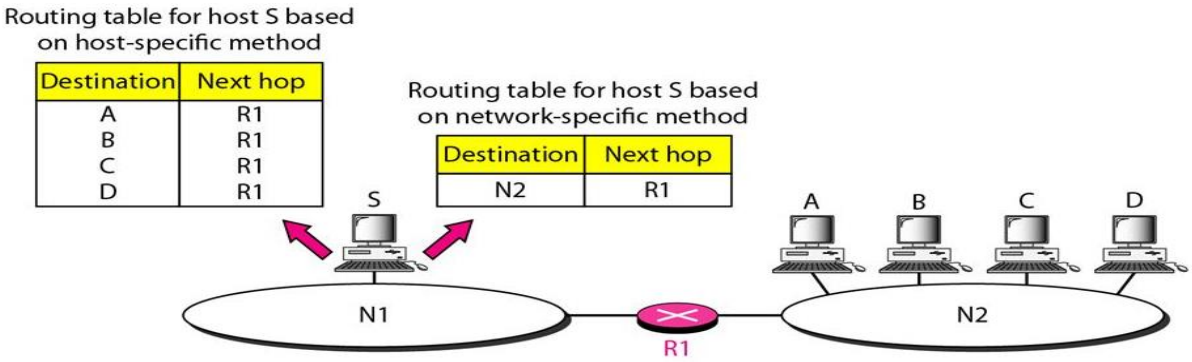


Figure: Host-specific versus network-specific method

Default Method

- Another technique to simplify routing is called the default method.
- In Figure host A is connected to a network with two routers.
- Router Rl routes the packets to hosts connected to network N2. However, for the rest of the Internet, router R2 is used.
- So instead of listing all networks in the entire Internet, host A can just have one entry called the *default* (normally defined as network address 0.0.0.0).

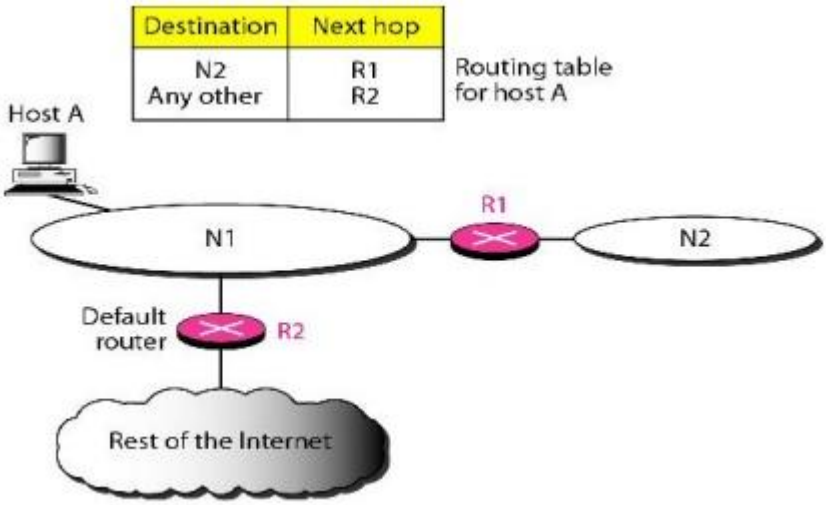
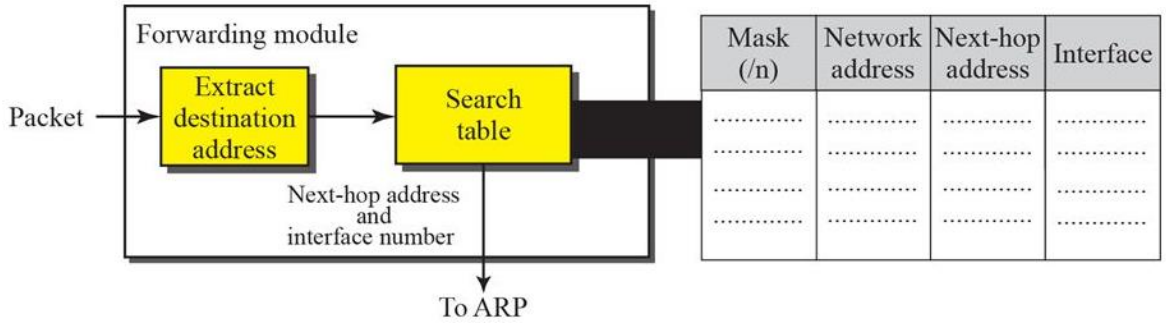


Figure: Default method

Forwarding Process

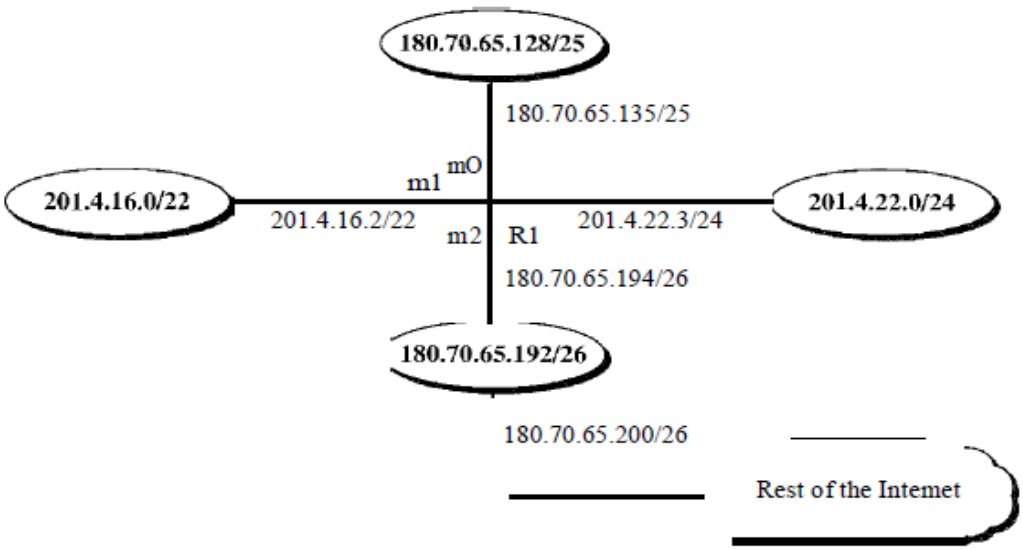
- We assume that hosts and routers use classless addressing because classful addressing can be treated as a special case of classless addressing. In classless addressing, the routing table needs to have one row of information for each block involved. The table needs to be searched based on the network address (first address in the block). Unfortunately, the destination address in the packet gives no clue about the network address. To solve the problem, we need to include the mask *(/n)* in the table; we need to have an extra column that includes the mask for the corresponding block. Figure shows a simple forwarding module for classless addressing.

- Note that we need at least four columns in our routing table; usually there are more.
- **In classless addressing, we need at least four colwnns in a routing table.**



Example :

Make a routing table for router Rl, using the configuration in Figure.



Solution:

Table *Routing table for router RI in Figure.*

Mask	*Network Address*	*Next Hop*	*Interface*
/26	180.70.65.192	-	m2
/25	180.70.65.128	-	mO
/24	201.4.22.0	-	m3
/22	201.4.16.0	....	m1
Any	Any	180.70.65.200	m2

Routing Table

- A **static routing table** contains information entered manually.
- The administrator enters the route for each destination into the table.
- When a table is created, it cannot update automatically when there is a change in the Internet. The table must be manually altered by the administrator.
- **A static routing table can be used in a small internet that does not change very often**, or in an experimental internet for troubleshooting.
- A **dynamic routing table** is updated periodically by using one of the dynamic routing protocols such as RIP, OSPF, or BGP.
- Whenever there is a change in the Internet, such as a shutdown of a router or breaking of a link, the dynamic routing protocols update all the tables in the routers (and eventually in the host) automatically.
- The routers in **a big internet such as the Internet need to be updated dynamically** for efficient delivery of the IP packets

Format

Mask	Network address	Next-hop address	Interface	Flags	Reference count	Use
.....	.....	.....	.....	.....	.....	.....

Common fields in a routing table

Mask. This field defines the mask applied for the entry.

Network address. This field defines the network address to which the packet is finally delivered. In the case of host-specific routing, this field defines the address of the destination host.

Next-hop address. This field defines the address of the next-hop router to which the packet is delivered.

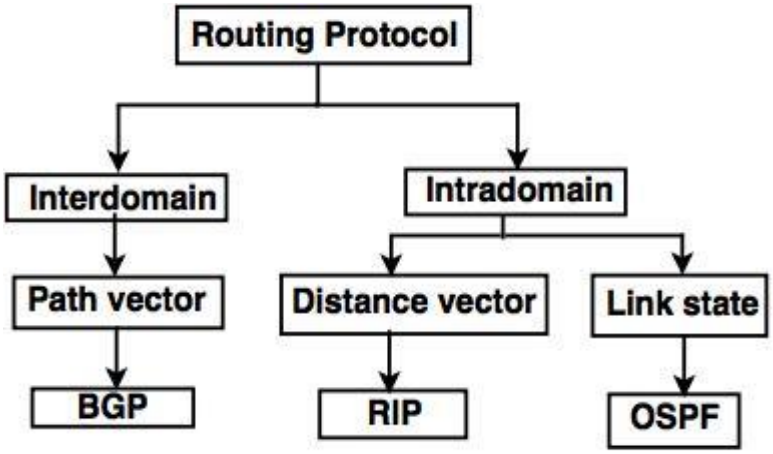
Interface. This field shows the name of the interface.

Flags. This field defines up to five flags. Flags are on/off switches that signify either presence or absence. The five flags are U (up), G (gateway), H (host-specific), D (added by redirection), and M (modified by redirection).

Reference count. This field gives the number of users of this route at the moment.

For example, if five people at the same time are connecting to the same host from this router, the value of this column is 5.

Use. This field shows the number of packets transmitted through this router for the corresponding destination.

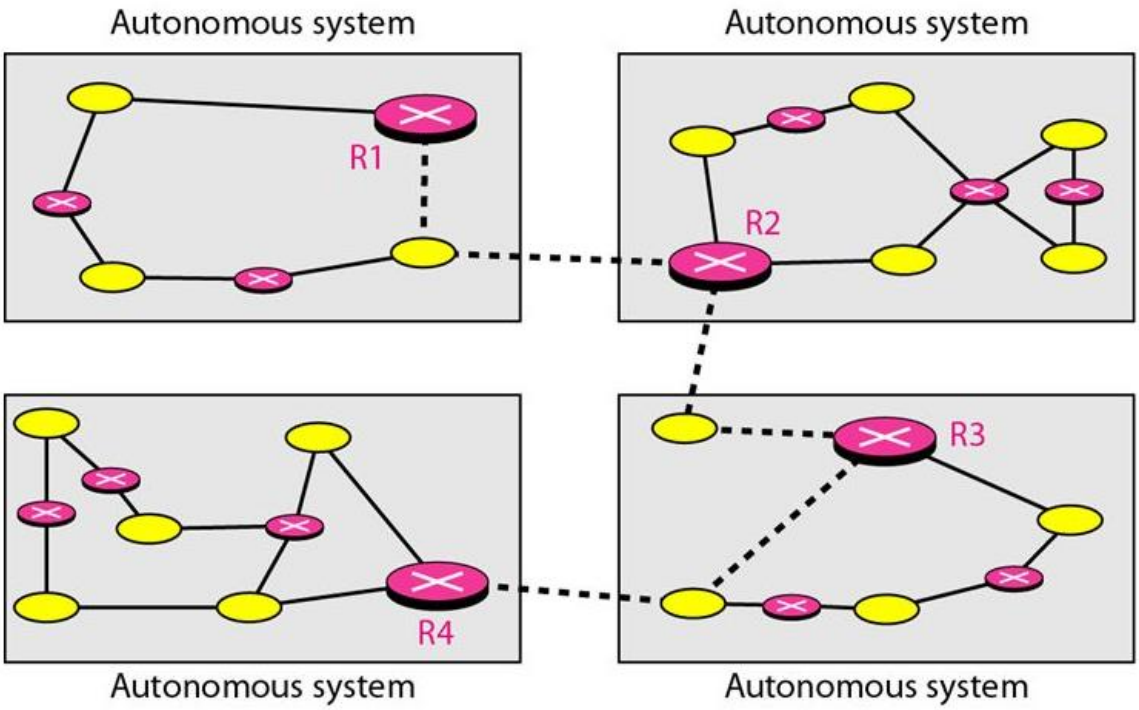


Classification of routing protocol

Intradomain and Interdomain Routing

- Today, an internet can be so large that one routing protocol cannot handle the task of updating the routing tables of all routers.
- For this reason, an internet is divided into **autonomous systems.**
- **An autonomous system (AS)** is a group of networks and routers under the authority of a single administration.
- **Routing inside an autonomous system is referred to as intradomain routing.**
- **Routing between autonomous systems is referred to as interdomain routing.**
- Each autonomous system can choose **one or more intradomain routing protocols** to handle routing inside the autonomous system.

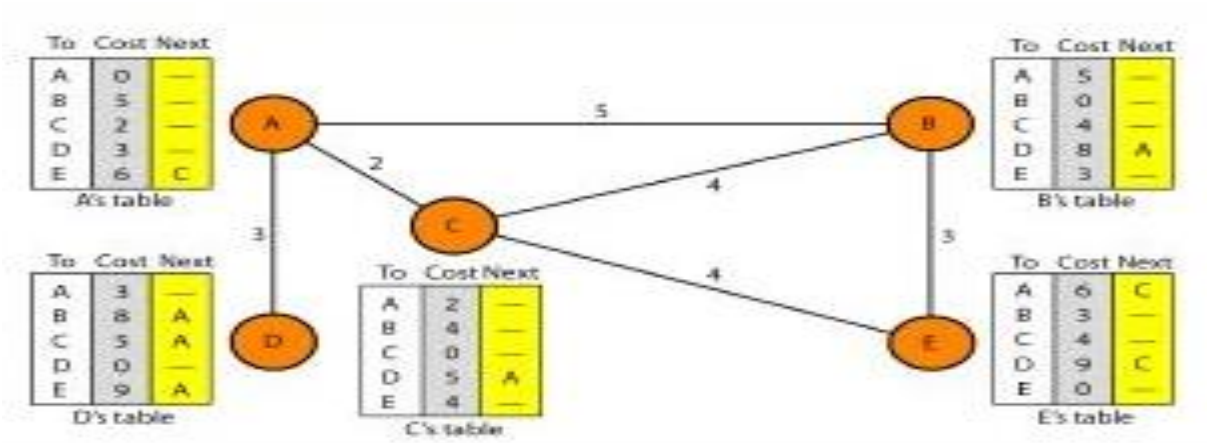
- However, **only one interdomain routing protocol** handles routing between autonomous systems.



- Several intradomain and interdomain routing protocols are in use.
- **Two intradomain** routing protocols: **distance vector and link state**.
- **One interdomain** routing protocol: **path vector**
- **Routing Information Protocol (RIP)** is an implementation of the **distance vector** protocol.
- **Open Shortest Path First (OSPF)** is an implementation of the **link state** protocol.
- **Border Gateway Protocol (BGP)** is an implementation of the **path vector** protocol.

Distance Vector Routing

- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this protocol, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).
- We can think of nodes as the cities in an area and the lines as the roads connecting them.



- **In distance vector routing, each node shares its routing table with its immediate neighbours periodically and when there is a change.**
- **Distance Vector Routing is a dynamic routing algorithm**

Step-01:

Each router prepares its routing table. By their local knowledge. each router knows about-

- All the routers present in the network
- Distance to its neighbouring routers

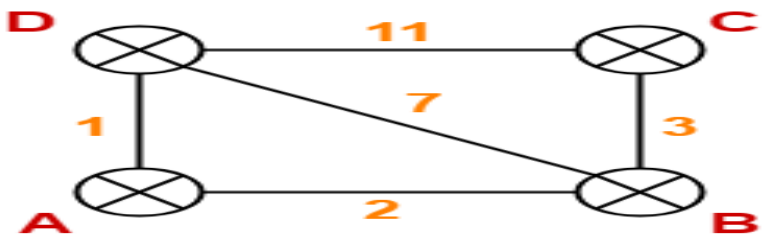
Step-02:

- Each router exchanges its distance vector with its neighbouring routers.
- Each router prepares a new routing table using the distance vectors it has obtained from its neighbours.
- This step is repeated for (n-2) times if there are n routers in the network.
- After this, routing tables converge / become stable.

Distance Vector Routing Example-

Consider-

- There is a network consisting of 4 routers.
- The weights are mentioned on the edges.
- Weights could be distances or costs or delays.



Step-01:

Each router prepares its routing table using its local knowledge.

Routing table prepared by each router is shown below-

At Router A-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	–
D	1	D

At Router B-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C

D	7	D

At Router C-

Destination	**Distance**	**Next Hop**
A	∞	–
B	3	B
C	0	C
D	11	D

At Router D-

Destination	**Distance**	**Next Hop**
A	1	A
B	7	B
C	11	C
D	0	D

Step-02:

- Each router exchanges its distance vector obtained in Step-01 with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.

This is shown below-

At Router A-

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B

2
0
3
7

Cost(A→B) = 2

From D

1
7
11
0

Cost(A→D) = 1

Destination	Distance	Next hop
A	0	A
B		
C		
D		

New Routing Table at Router A

- Cost of reaching destination B from router A = min { 2+0 , 1+7 } = 2 via B.
- Cost of reaching destination C from router A = min { 2+3 , 1+11 } = 5 via B.
- Cost of reaching destination D from router A = min { 2+7 , 1+0 } = 1 via D.

Explanation For Destination B

- Router A can reach the destination router B via its neighbor B or neighbor D.
- It chooses the path which gives the minimum cost.
- Cost of reaching router B from router A via neighbor B = Cost (A→B) + Cost (B→B)= **2 + 0** = 2
- Cost of reaching router B from router A via neighbor D = Cost (A→D) + Cost (D→B) = **1 + 7** = 8
- Since the cost is minimum via neighbor B, so router A chooses the path via B.

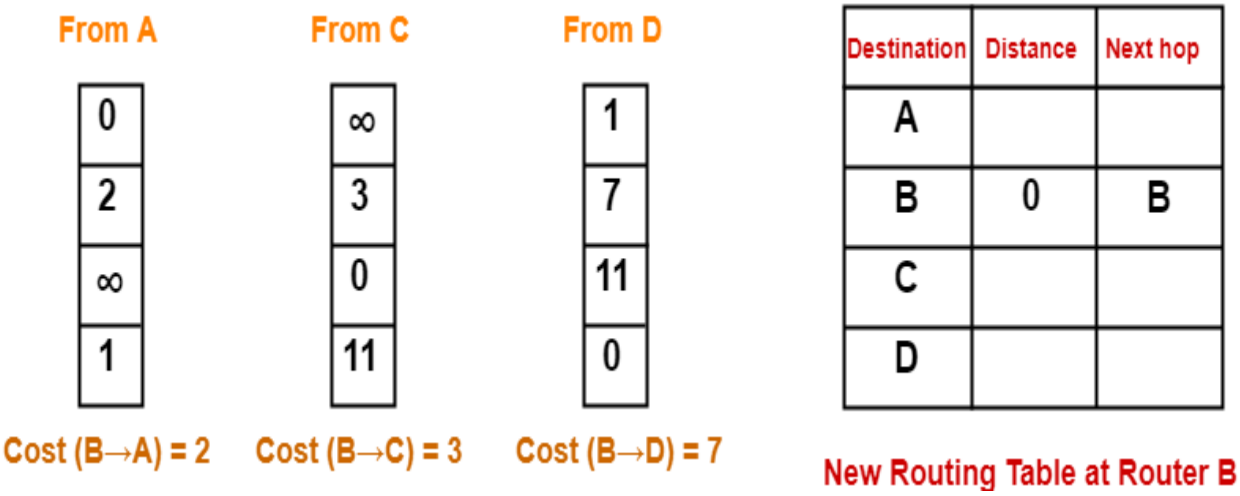
- It creates an entry (2, B) for destination B in its new routing table.
- Similarly, we calculate the shortest path distance to each destination router at every router.

Thus, the new routing table at router A is-

Destination	**Distance**	**Next Hop**
A	0	A
B	2	B
C	5	B
D	1	D

<u>At Router B-</u>

- Router B receives distance vectors from its neighbours A, C and D.
- Router B prepares a new routing table as-



- Cost of reaching destination A from router B = min { 2+0 , 3+∞ , 7+1 } = 2 via A.
- Cost of reaching destination C from router B = min { 2+∞ , 3+0 , 7+11 } = 3 via C.
- Cost of reaching destination D from router B = min { 2+1 , 3+11 , 7+0 } = 3 via A.

Thus, the new routing table at router B is-

Destination	**Distance**	**Next Hop**
A	2	A
B	0	B
C	3	C
D	3	A

<u>At Router C-</u>

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-

From B

2
0
3
7

Cost (C→B) = 3

From D

1
7
11
0

Cost (C→D) = 11

Destination	Distance	Next hop
A		
B		
C	0	C
D		

New Routing Table at Router C

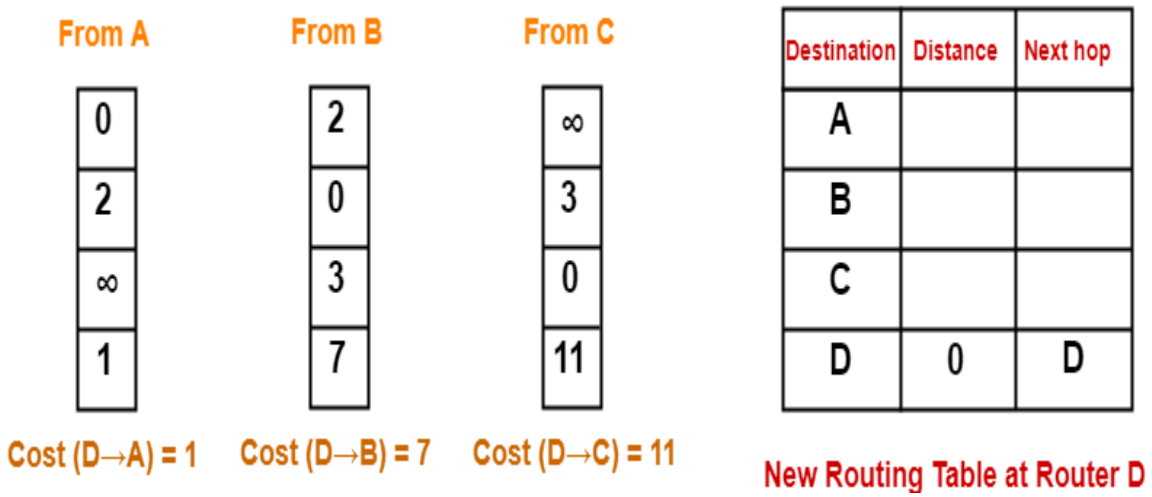
- Cost of reaching destination A from router C = min { 3+2 , 11+1 } = 5 via B.
- Cost of reaching destination B from router C = min { 3+0 , 11+7 } = 3 via B.
- Cost of reaching destination D from router C = min { 3+7 , 11+0 } = 10 via B.

Thus, the new routing table at router C is-

Destination	**Distance**	**Next Hop**
A	5	B
B	3	B
C	0	C
D	10	B

<u>At Router D-</u>

- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-



- Cost of reaching destination A from router D = min { 1+0 , 7+2 , 11+∞ } = 1 via A.
- Cost of reaching destination B from router D = min { 1+2 , 7+0 , 11+3 } = 3 via A.
- Cost of reaching destination C from router D = min { 1+∞ , 7+3 , 11+0 } = 10 via B.

Thus, the new routing table at router D is-

Destination	**Distance**	**Next Hop**
A	1	A
B	3	A
C	10	B
D	0	D

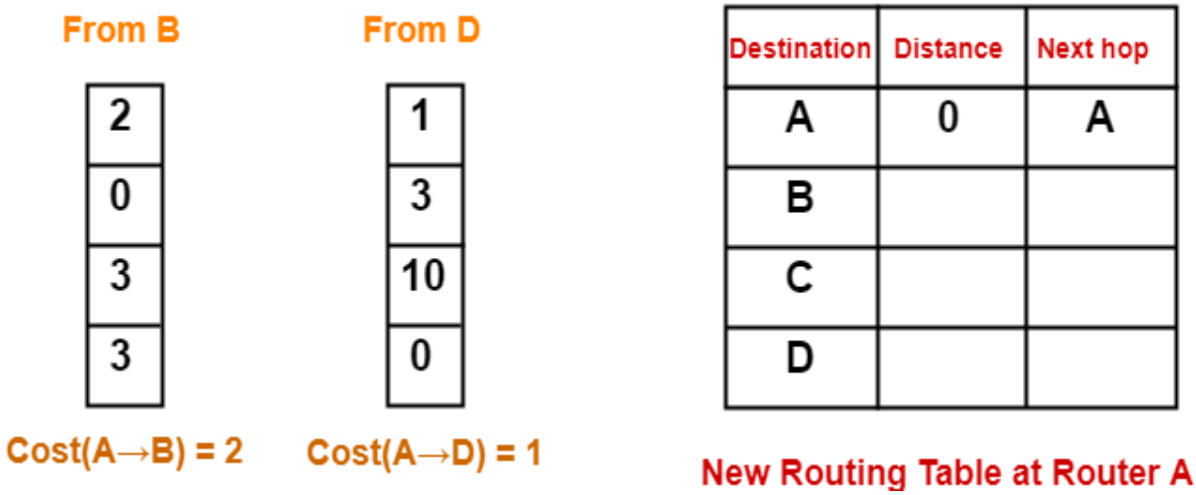
Step-03:

- Each router exchanges its distance vector obtained in Step-02 with its neighbouring routers.
- After exchanging the distance vectors, each router prepares a new routing table.

This is shown below-

At Router A-

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-



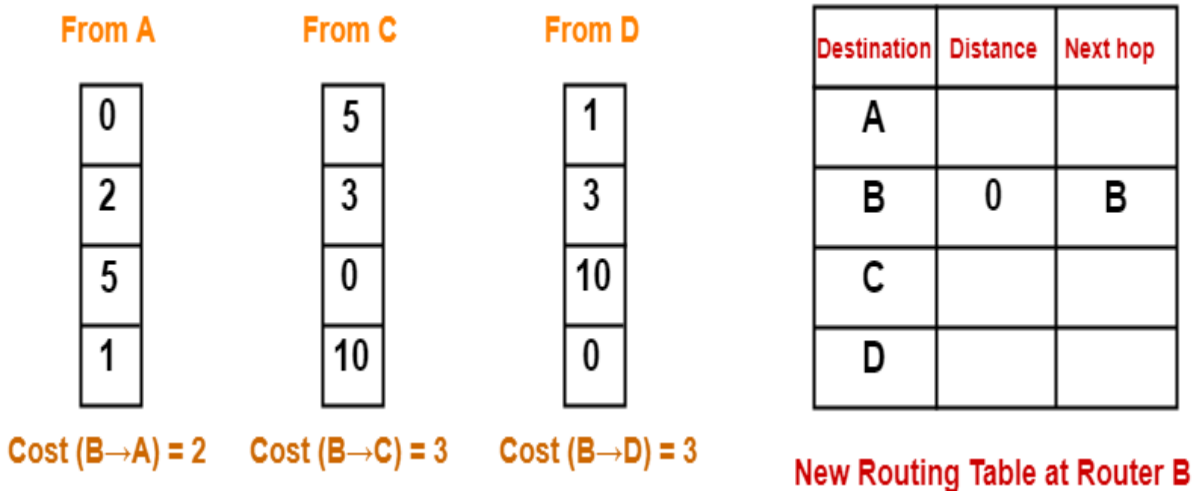
- Cost of reaching destination B from router A = min { 2+0 , 1+3 } = 2 via B.
- Cost of reaching destination C from router A = min { 2+3 , 1+10 } = 5 via B.
- Cost of reaching destination D from router A = min { 2+3 , 1+0 } = 1 via D.

Thus, the new routing table at router A is-

Destination	**Distance**	**Next Hop**
A	0	A
B	2	B
C	5	B
D	1	D

<u>At Router B-</u>

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-



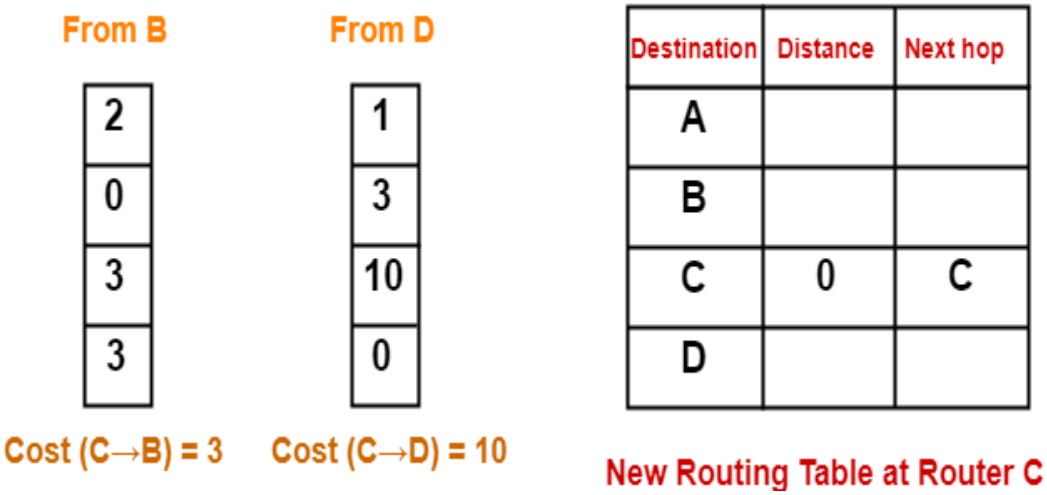
- Cost of reaching destination A from router B = min { 2+0 , 3+5 , 3+1 } = 2 via A.
- Cost of reaching destination C from router B = min { 2+5 , 3+0 , 3+10 } = 3 via C.
- Cost of reaching destination D from router B = min { 2+1 , 3+10 , 3+0 } = 3 via A.

Thus, the new routing table at router B is-

Destination	**Distance**	**Next Hop**
A	2	A
B	0	B
C	3	C
D	3	A

<u>At Router C-</u>

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-



- Cost of reaching destination A from router C = min { 3+2 , 10+1 } = 5 via B.
- Cost of reaching destination B from router C = min { 3+0 , 10+3 } = 3 via B.
- Cost of reaching destination D from router C = min { 3+3 , 10+0 } = 6 via B.

Thus, the new routing table at router C is-

Destination	**Distance**	**Next Hop**
A	5	B
B	3	B
C	0	C
D	6	B

<u>At Router D-</u>

- Router D receives distance vectors from its neighbours A, B and C.
- Router D prepares a new routing table as-

- Cost of reaching destination A from router D = min { 1+0 , 3+2 , 10+5 } = 1 via A.
- Cost of reaching destination B from router D = min { 1+2 , 3+0 , 10+3 } = 3 via A.
- Cost of reaching destination C from router D = min { 1+5 , 3+3 , 10+0 } = 6 via A.

Thus, the new routing table at router D is-

Destination	**Distance**	**Next Hop**
A	1	A
B	3	A
C	6	A
D	0	D

These will be the final routing tables at each router.

Identifying Unused Links-

After routing tables converge (becomes stable),

- Some of the links connecting the routers may never be used.
- In the above example, we can identify the unused links as-

We have-

- The value of next hop in the final routing table of router A suggests that only edges AB and AD are used.
- The value of next hop in the final routing table of router B suggests that only edges BA and BC are used.
- The value of next hop in the final routing table of router C suggests that only edge CB is used.
- The value of next hop in the final routing table of router D suggests that only edge DA is used.

Thus, edges BD and CD are never used.

In Distance Vector Routing,

- Only distance vectors are exchanged.
- "Next hop "values are not exchanged.
- This is because it results in exchanging the large amount of data which consumes more bandwidth.

While preparing a new routing table-

- A router takes into consideration only the distance vectors it has obtained from its neighbouring routers.
- It does not take into consideration its old routing table.

The algorithm is called so because-

- It involves exchanging of distance vectors between the routers.
- Distance vector is nothing but an array of distances.

- **The algorithm keeps on repeating periodically and never stops.**
- **This is to update the shortest path in case any link goes down or topology changes.**
- **Routing tables are prepared total (n-1) times if there are n routers in the given network.**
- **This is because shortest path between any 2 nodes contains at most n-1 edges if there are n nodes in the graph.**
- **Distance Vector Routing suffers from count to infinity problem.**
- **Distance Vector Routing uses UDP at transport layer**

Address Resolution Protocol (ARP) and its types

- Address Resolution Protocol (ARP) is a communication protocol used to find the MAC (Media Access Control) address of a device from its IP address.
- This protocol is used when a device wants to communicate with another device on a Local Area Network or Ethernet.

Types of ARP

- There are four types of Address Resolution Protocol, which is given below:
    - Proxy ARP
    - Gratuitous ARP
    - Reverse ARP (RARP)
    - Inverse ARP

Proxy ARP - Proxy ARP is a method through which a Layer 3 devices may respond to ARP requests for a target that is in a different network from the sender. The Proxy ARP configured router responds to the ARP and map the MAC address of the router with the target IP address and fool the sender that it is reached at its destination.

At the backend, the proxy router sends its packets to the appropriate destination because the packets contain the necessary information.

Example - If Host A wants to transmit data to Host B, which is on the different network, then Host A sends an ARP request message to receive a MAC address for Host B. The router responds to Host A with its own MAC address pretend itself as a destination. When the data is transmitted to the destination by Host A, it will send to the gateway so that it sends to Host B. This is known as proxy ARP.

Gratuitous ARP - Gratuitous ARP is an ARP request of the host that helps to identify the duplicate IP address. It is a broadcast request for the IP address of the router. If an ARP request is sent by a switch or router to get its IP address and no ARP responses are received, so all other nodes cannot use the IP address allocated to that switch or router. Yet if a router or switch sends an ARP request for its IP address and receives an ARP response, another node uses the IP address allocated to the switch or router.

There are some primary use cases of gratuitous ARP that are given below:

- The gratuitous ARP is used to update the ARP table of other devices.
- It also checks whether the host is using the original IP address or a duplicate one.

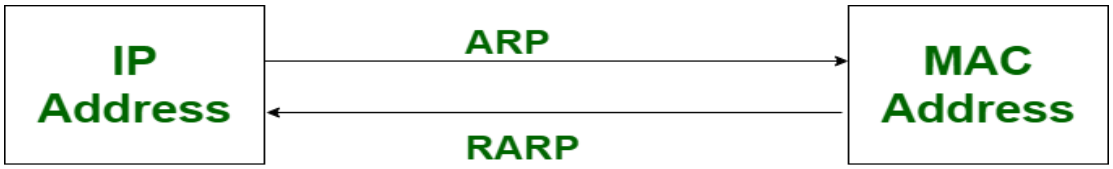
Reverse ARP (RARP) - It is a networking protocol used by the client system in a local area network (LAN) to request its IPv4 address from the ARP gateway router table. A table is created by the network administrator in the gateway-router that is used to find out the MAC address to the corresponding IP address.

When a new system is set up or any machine that has no memory to store the IP address, then the user has to find the IP address of the device. The device sends a RARP broadcast packet, including its own MAC address in the address field of both the sender and the receiver hardware. A host installed inside of the local network called the RARP-server is prepared to respond to such type of broadcast packet. The RARP server is then trying to locate a mapping

table entry in the IP to MAC address. If any entry matches the item in the table, then the RARP server sends the response packet along with the IP address to the requesting computer.

Inverse ARP (InARP) - Inverse ARP is inverse of the ARP, and it is used to find the IP addresses of the nodes from the data link layer addresses. These are mainly used for the frame relays, and ATM networks, where Layer 2 virtual circuit addressing are often acquired from Layer 2 signaling. When using these virtual circuits, the relevant Layer 3 addresses are available.

ARP conversions Layer 3 addresses to Layer 2 addresses. However, its opposite address can be defined by InARP. The InARP has a similar packet format as ARP, but operational codes are different.



S.NO	**ARP**	**RARP**
1.	ARP stands for Address Resolution Protocol.	Whereas RARP stands for Reverse Address Resolution Protocol.
2.	Through ARP, (32-bit) IP address mapped into (48-bit) MAC address.	Whereas through RARP, (48-bit) MAC address of 48 bits mapped into (32-bit) IP address.
3.	In ARP, broadcast MAC address is used.	While in RARP, broadcast IP address is used.
4.	In ARP, ARP table is managed or maintained by local host.	While in RARP, RARP table is managed or maintained by RARP server.
5.	In Address Resolution Protocol, Receiver's MAC address is fetched.	While in RARP, IP address is fetched.
6.	In ARP, ARP table uses ARP reply for its updation.	While in RARP, RARP table uses RARP reply for configuration of IP addresses .

7.	Hosts and routers uses ARP for knowing the MAC address of other hosts and routers in the networks.	While RARP is used by small users having less facilities.

Internet Control Message Protocol (ICMP)

- The Internet Control Message Protocol (ICMP) is a network layer protocol.
- It is used by network devices to diagnose network communication issues.
- ICMP is mainly used to determine whether or not data is reaching its intended destination in a timely manner.
- Commonly, the ICMP protocol is used on network devices, such as routers.
- ICMP is crucial for error reporting and testing, but it can also be used in distributed denial-of-service (DDoS) attacks.

Type(8 bit)	Code(8 bit)	CheckSum(16 bit)
Extended Header(32 bit)		
Data/Payload(Variable Length)		

How does ICMP work?

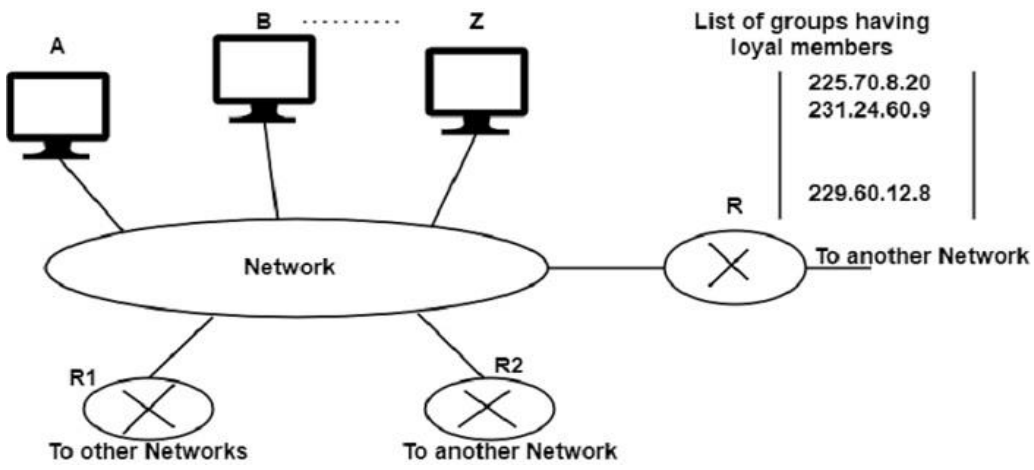
- ICMP is not associated with a transport layer protocol such as TCP or UDP.
- ICMP is a connectionless protocol i.e one device does not need to open a connection with another device before sending an ICMP message.
- Normal IP traffic is sent using TCP, which means any two devices that exchange data will first carry out a TCP handshake to ensure both devices are ready to receive data.
- ICMP does not open a connection in this way. The ICMP protocol also does not allow for targeting a specific port on a device.

IGMP (Internet Group Management Protocol)

- **IGMP** is acronym for **Internet Group Management Protocol**.
- IGMP is a communication protocol used by hosts and adjacent routers for multicasting communication with IP networks and uses the resources efficiently to transmit the message/data packets.
- IGMP is used on network layer protocol.
- It is a protocol that allows several devices to share one IP address so they can all receive the same data.
- Multicast communication can have single or multiple senders and receivers and thus, IGMP can be used in streaming videos, gaming or web conferencing tools.
- This protocol is used on IPv4 networks and for using this on IPv6, multicasting is managed by Multicast Listener Discovery (MLD).
- IGMP is not a multicasting routing protocol; it is a protocol that manages group membership. In any network, there are one or more multicast routers that distribute multicast packets to hosts or other routers.
- A multicast router may receive thousands of multicast packets every day for different groups. If a router has no knowledge about the membership status of the hosts, it must broadcast all these packets.
- This creates a lot of traffic and consumes bandwidth. A better solution is to keep a list of groups in the network for which there is at least one loyal member.
- IGMP helps the multicast router create and update this list.

Operations of IGMP

- IGMP operates locally. A multicast router connected to a network has a list of multicast addresses of the groups with at least one loyal member in that network, as shown in the figure below −



- For each group, there is one router that has the duty of distributing the multicast packets destined for that group. This means that if there are three multicast routers connected to a network, their lists of group ids are mutually exclusive.
- A host or multicast router can have membership in a group.
- When a host has membership, it means that one of its processes (an application program) receives multicast packets from some group.
- When a router has membership, it means that a network connection to one of its other interfaces receives these multicast packets. We say that the host or the router has an interest in the group.
- In both cases, the host and the router keep a list of group ids and relay their interest to the distributing router.
- There are two other multicast routers (R1 and R2) that, depending on the group list maintained by router R, could be the recipients of router R in this network.
- Routers RI and R2 may be distributors for some of these groups in other networks but not on this network.

Advantages:

- IGMP communication protocol efficiently transmits the multicast data to the receivers and so, no junk packets are transmitted to the host which shows optimized performance.
- Bandwidth is consumed totally as all the shared links are connected.
- Hosts can leave a multicast group and join another.

Disadvantages:

- It does not provide good efficiency in filtering and security.
- Due to lack of TCP, network congestion can occur.
- IGMP is vulnerable to some attacks such as DOS attack (Denial-Of-Service).

RIP Protocol

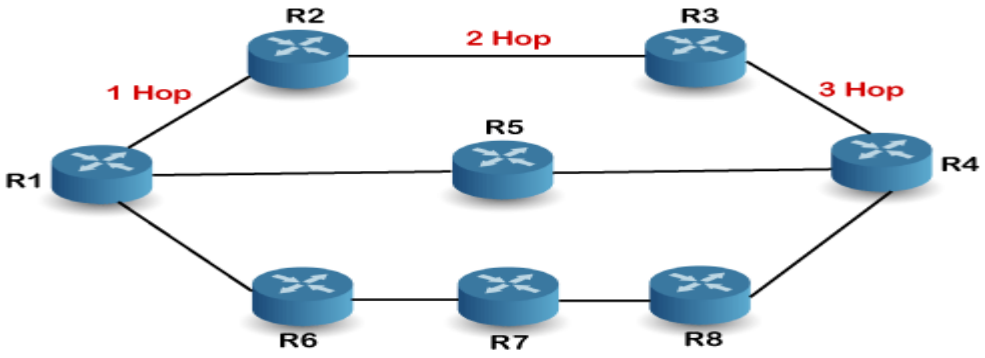
- RIP stands for Routing Information Protocol.
- RIP is an intra-domain routing protocol used within an autonomous system. Here, intra-domain means routing the packets in a defined domain, for example, web browsing within an institutional area.

Before understanding the structure of the packet, we first look at the following points:

- RIP is based on the distance vector-based strategy, so we consider the entire structure as a graph where nodes are the routers, and the links are the networks.
- In a routing table, the first column is the destination, or we can say that it is a network address.
- The cost metric is the number of hops to reach the destination. The number of hops available in a network would be the cost. The hop count is the number of networks required to reach the destination.
- In RIP, infinity is defined as 16, which means that the RIP is useful for smaller networks or small autonomous systems. The maximum number of hops that RIP can contain is 15 hops, i.e., it should not have more than 15 hops as 16 is infinity.
- The next column contains the address of the router to which the packet is to be sent to reach the destination.

How is hop count determined?

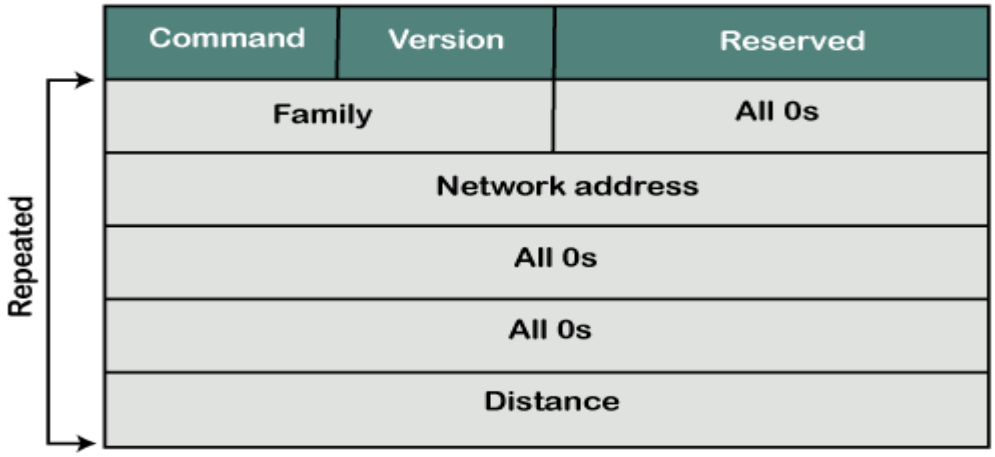
When the router sends the packet to the network segment, then it is counted as a single hop.



In the above figure, when the router 1 forwards the packet to the router 2 then it will count as 1 hop count. Similarly, when the router 2 forwards the packet to the router 3 then it will count as 2 hop count, and when the router 3 forwards the packet to router 4, it will count as 3 hop count. In the same way, RIP can support maximum upto 15 hops, which means that the 16 routers can be configured in a RIP.

RIP Message Format

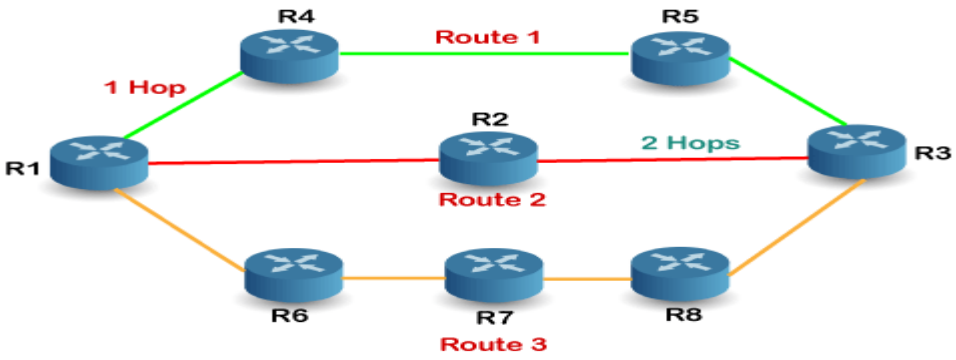
The message format is used to share information among different routers. The RIP contains the following fields in a message:



- **Command:** It is an 8-bit field that is used for request or reply. The value of the request is 1, and the value of the reply is 2.

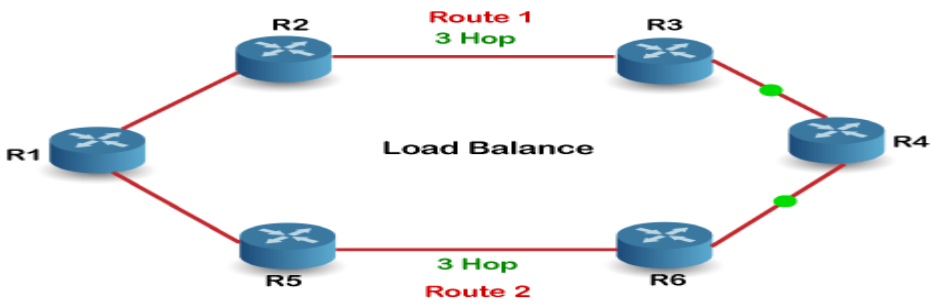
- **Version:** Here, version means that which version of the protocol we are using. Suppose we are using the protocol of version1, then we put the 1 in this field.
- **Reserved:** This is a reserved field, so it is filled with zeroes.
- **Family:** It is a 16-bit field. As we are using the TCP/IP family, so we put 2 value in this field.
- **Network Address:** It is defined as 14 bytes field. If we use the IPv4 version, then we use 4 bytes, and the other 10 bytes are all zeroes.
- **Distance:** The distance field specifies the hop count, i.e., the number of hops used to reach the destination.

How does the RIP work?



- If there are 8 routers in a network where Router 1 wants to send the data to Router 3. If the network is configured with RIP, it will choose the route which has the least number of hops. There are three routes in the above network, i.e., Route 1, Route 2, and Route 3. The Route 2 contains the least number of hops, i.e., 2 where Route 1 contains 3 hops, and Route 3 contains 4 hops, so RIP will choose Route 2.

Let's look at another example.

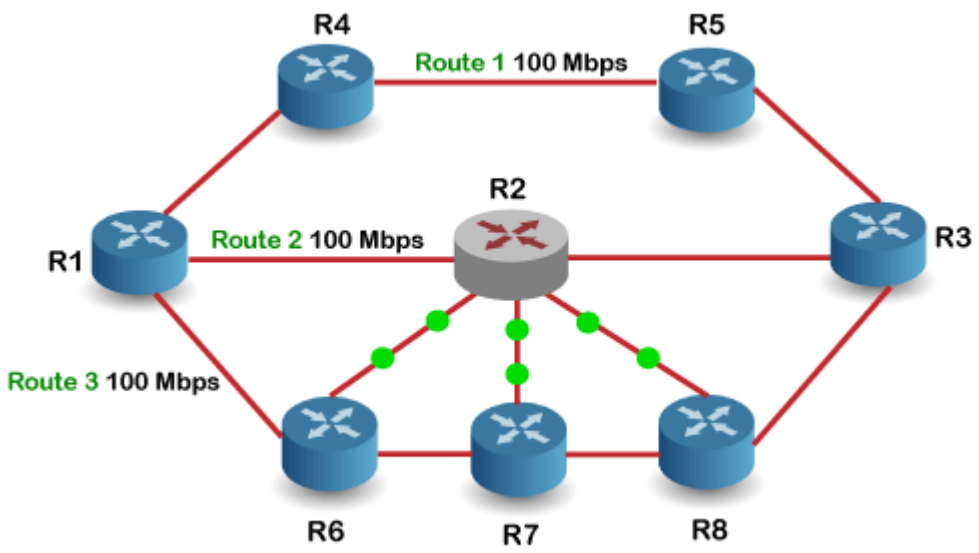


- Suppose R1 wants to send the data to R4. There are two possible routes to send data from r1 to r2. As both the routes contain the same number of hops, i.e., 3, so RIP will send the data to both the routes simultaneously. This way, it manages the load balancing, and data reach the destination a bit faster.

Disadvantages of RIP

The following are the disadvantages of RIP:

- In RIP, the route is chosen based on the hop count metric. If another route of better bandwidth is available, then that route would not be chosen. Let's understand this scenario through an example.



We can observe that Route 2 is chosen in the above figure as it has the least hop count. The Route 1 is free and data can be reached more faster; instead of this, data is sent to the Route 2 that makes the Route 2 slower due to the heavy traffic. This is one of the biggest disadvantages of RIP.

- The RIP is a classful routing protocol, so it does not support the VLSM (Variable Length Subnet Mask). The classful routing protocol is a protocol that does not include the subnet mask information in the routing updates.
- It broadcasts the routing updates to the entire network that creates a lot of traffic. In RIP, the routing table updates every 30 seconds. Whenever the updates occur, it sends

the copy of the update to all the neighbors except the one that has caused the update. The sending of updates to all the neighbors creates a lot of traffic. This rule is known as a split-horizon rule.

- It faces a problem of Slow convergence. Whenever the router or link fails, then it often takes minutes to stabilize or take an alternative route; This problem is known as Slow convergence.
- RIP supports maximum 15 hops which means that the maximum 16 hops can be configured in a RIP
- The Administrative distance value is 120 (Ad value). If the Ad value is less, then the protocol is more reliable than the protocol with more Ad value.
- The RIP protocol has the highest Ad value, so it is not as reliable as the other routing protocols.

How RIP updates its Routing table

The following timers are used to update the routing table:

- **RIP update timer : 30 sec**

The routers configured with RIP send their updates to all the neighboring routers every 30 seconds.

- **RIP Invalid timer : 180 sec**

The RIP invalid timer is 180 seconds, which means that if the router is disconnected from the network or some link goes down, then the neighbor router will wait for 180 seconds to take the update. If it does not receive the update within 180 seconds, then it will mark the particular route as not reachable.

- **RIP Flush timer : 240 sec**

The RIP flush timer is 240 second which is almost equal to 4 min means that if the router does not receive the update within 240 seconds then the neighbor route will remove that particular route from the routing table which is a very slow process as 4 minutes is a long time to wait.

Advantages of RIP

The following are the advantages of a RIP protocol:

- It is easy to configure
- It has less complexity
- The CPU utilization is less.

Link State Routing

Link state routing is a technique in which each router shares the knowledge of its neighborhood with every other router in the internetwork.

The three keys to understand the Link State Routing algorithm:

- **Knowledge about the neighborhood:** Instead of sending its routing table, a router sends the information about its neighborhood only. A router broadcast its identities and cost of the directly attached links to other routers.
- **Flooding:** Each router sends the information to every other router on the internetwork except its neighbors. This process is known as Flooding. Every router that receives the packet sends the copies to all its neighbors. Finally, each and every router receives a copy of the same information.
- **Information sharing:** A router sends the information to every other router only when the change occurs in the information.

Link State Routing has two phases:

Reliable Flooding

- **Initial state:** Each node knows the cost of its neighbors.
- **Final state:** Each node knows the entire graph.

Route Calculation

Each node uses Dijkstra's algorithm on the graph to calculate the optimal routes to all nodes.

- The Link state routing algorithm is also known as Dijkstra's algorithm which is used to find the shortest path from one node to every other node in the network.
- The Dijkstra's algorithm is an iterative, and it has the property that after k^{th} iteration of the algorithm, the least cost paths are well known for k destination nodes.

Let's describe some notations:

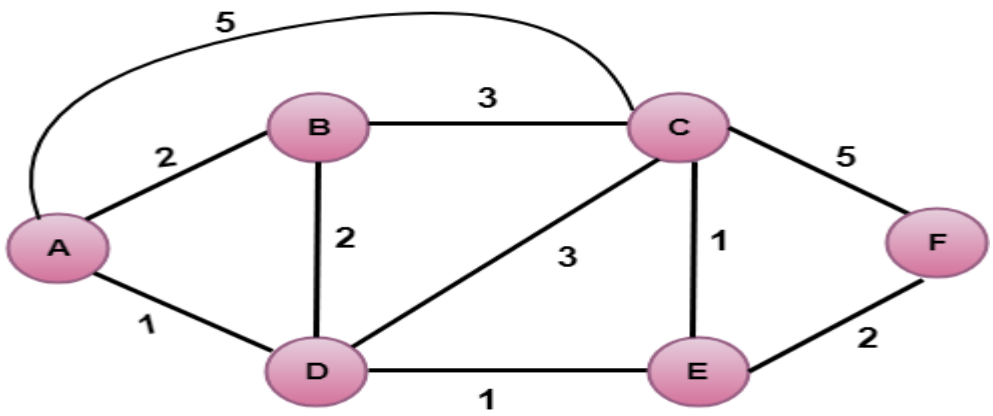
- **c(i , j):** Link cost from node i to node j. If i and j nodes are not directly linked, then c(i , j) = ∞.
- **D(v):** It defines the cost of the path from source code to destination v that has the least cost currently.
- **P(v):** It defines the previous node (neighbor of v) along with current least cost path from source to v.
- **N:** It is the total number of nodes available in the network.

Algorithm

```
Initialization
N = {A}      // A is a root node.
for all nodes v
if v adjacent to A
then D(v) = c(A,v)
else D(v) = infinity
loop
find w not in N such that D(w) is a minimum.
Add w to N
Update D(v) for all v adjacent to w and not in N:
D(v) = min(D(v) , D(w) + c(w,v))
Until all nodes in N
```

In the above algorithm, an initialization step is followed by the loop. The number of times the loop is executed is equal to the total number of nodes available in the network.

Let's understand through an example:



In the above figure, source vertex is A.

Step 1:

The first step is an initialization step. The currently known least cost path from A to its directly attached neighbors, B, C, D are 2,5,1 respectively. The cost from A to B is set to 2, from A to D is set to 1 and from A to C is set to 5. The cost from A to E and F are set to infinity as they are not directly linked to A.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞

Step 2:

In the above table, we observe that vertex D contains the least cost path in step 1. Therefore, it is added in N. Now, we need to determine a least-cost path through D vertex.

a) Calculating shortest path from A to B

v = B, w = D

D(B) = min(D(B) , D(D) + c(D,B))

= min(2, 1+2)>

= min(2, 3)

The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating shortest path from A to C

v = C, w = D

D(B) = min(D(C) , D(D) + c(D,C))

= min(5, 1+3)

= min(5, 4)

The minimum value is 4. Therefore, the currently shortest path from A to C is 4.</p>

c) Calculating shortest path from A to E

v = E, w = D

D(B) = min(D(E) , D(D) + c(D,E))

= min(∞, 1+1)

= min(∞, 2)

The minimum value is 2. Therefore, the currently shortest path from A to E is 2.

Note: The vertex D has no direct link to vertex E. Therefore, the value of D(F) is infinity.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞

Step 3:

In the above table, we observe that both E and B have the least cost path in step 2. Let's consider the E vertex. Now, we determine the least cost path of remaining vertices through E.

a) Calculating the shortest path from A to B.

v = B, w = E

D(B) = min(D(B) , D(E) + c(E,B))

= min(2 , 2+ ∞)

= min(2, ∞)

The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating the shortest path from A to C.

v = C, w = E

D(B) = min(D(C) , D(E) + c(E,C))

= min(4 , 2+1)

= min(4,3)

The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

c) Calculating the shortest path from A to F.

v = F, w = E

D(B) = min(D(F) , D(E) + c(E,F))

= min(8 , 2+2)

= min(8 ,4)

The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E

Step 4:

In the above table, we observe that B vertex has the least cost path in step 3. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through B.

a) Calculating the shortest path from A to C.

> v = C, w = B
>
> D(B) = min(D(C) , D(B) + c(B,C))
>
> = min(3 , 2+3)
>
> = min(3,5)
>
> The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

b) Calculating the shortest path from A to F.

> v = F, w = B
>
> D(B) = min(D(F) , D(B) + c(B,F))
>
> = min(4, ∞)
>
> = min(4, ∞)
>
> The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E

Step 5:

In the above table, we observe that C vertex has the least cost path in step 4. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through C.

a) Calculating the shortest path from A to F.

> v = F, w = C
>
> D(B) = min(D(F) , D(C) + c(C,F))
>
> = min(4, 3+5)
>
> = min(4,8)
>
> The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E
5	ADEBC					4,E

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E

5	ADEBC					4,E
6	ADEBCF					

Final table

Disadvantage:

Heavy traffic is created in Line state routing due to Flooding. Flooding can cause an infinite looping, this problem can be solved by using Time-to-leave field

Congestion Control in Computer Networks

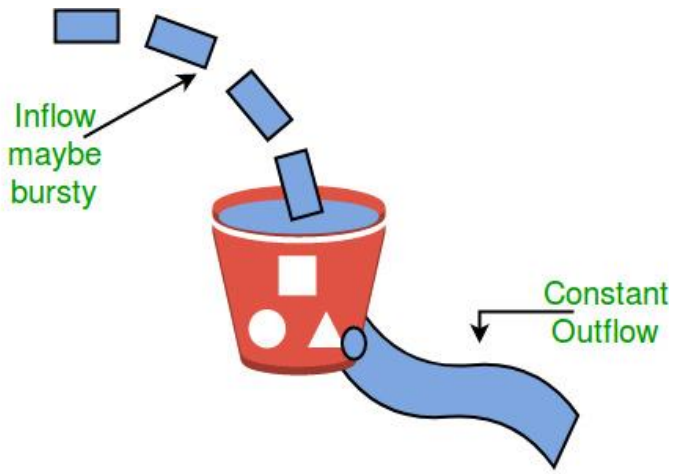
What is **congestion**?
A state occurring in network layer when the message traffic is so heavy that it slows down network response time.
Effects of Congestion

- As delay increases, performance decreases.
- If delay increases, retransmission occurs, making situation worse.

Congestion control algorithms

- **Leaky Bucket Algorithm**

Let us consider an example to understand
Imagine a bucket with a small hole in the bottom.No matter at what rate water enters the bucket, the outflow is at constant rate.When the bucket is full with water additional water entering spills over the sides and is lost.



Similarly, each network interface contains a leaky bucket and the following **steps** are involved in leaky bucket algorithm:

1. When host wants to send packet, packet is thrown into the bucket.
2. The bucket leaks at a constant rate, meaning the network interface transmits packets at a constant rate.
3. Bursty traffic is converted to a uniform traffic by the leaky bucket.
4. In practice the bucket is a finite queue that outputs at a finite rate.

- **Token bucket Algorithm**

Need of token bucket Algorithm:-

The leaky bucket algorithm enforces output pattern at the average rate, no matter how bursty the traffic is. So in order to deal with the bursty traffic we need a flexible algorithm so that the data is not lost. One such algorithm is token bucket algorithm.

Steps of this algorithm can be described as follows:

1. In regular intervals tokens are thrown into the bucket. *f*
2. The bucket has a maximum capacity. *f*
3. If there is a ready packet, a token is removed from the bucket, and the packet is sent.
4. If there is no token in the bucket, the packet cannot be sent.

Let's understand with an example,

In figure (A) we see a bucket holding three tokens, with five packets waiting to be transmitted. For a packet to be transmitted, it must capture and destroy one token. In figure (B) We see that three of the five packets have gotten through, but the other two are stuck waiting for more tokens to be generated.

Ways in which token bucket is superior to leaky bucket:

The leaky bucket algorithm controls the rate at which the packets are introduced in the

network, but it is very conservative in nature. Some flexibility is introduced in the token bucket algorithm. In the token bucket, algorithm tokens are generated at each tick (up to a certain limit). For an incoming packet to be transmitted, it must capture a token and the transmission takes place at the same rate. Hence some of the busty packets are transmitted at the same rate if tokens are available and thus introduces some amount of flexibility in the system.

Formula: $M * s = C + \rho * s$

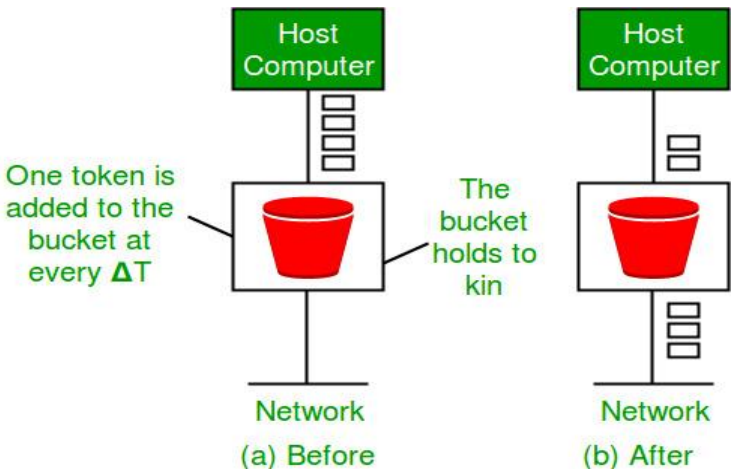
where S – is time taken

M – Maximum output rate

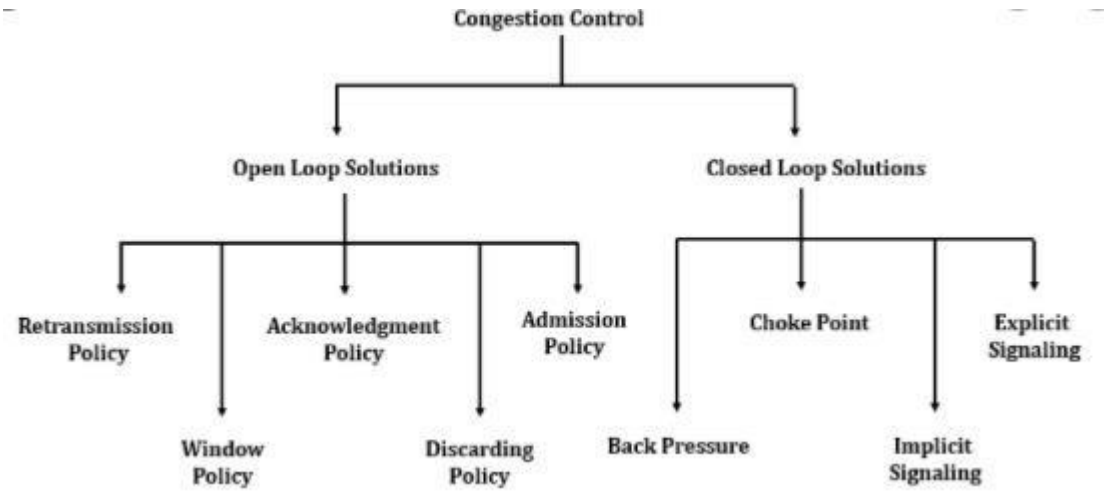
ρ – Token arrival rate

C – Capacity of the token bucket in byte

Let's understand with an example,



Congestion Control techniques in Computer Networks



- Congestion control refers to the techniques used to control or prevent congestion.
- Congestion control techniques can be broadly classified into two categories:

Open Loop Congestion Control

Open loop congestion control policies are applied to prevent congestion before it happens. The congestion control is handled either by the source or the destination.

Policies adopted by open loop congestion control –

1. **Retransmission Policy:**
   It is the policy in which retransmission of the packets are taken care of. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted. This transmission may increase the congestion in the network.
   To prevent congestion, retransmission timers must be designed to prevent congestion and also able to optimize efficiency.

2. **Window Policy:**
   The type of window at the sender's side may also affect the congestion. Several packets in the Go-back-n window are re-sent, although some packets may be received successfully at the receiver side. This duplication may increase the congestion in the network and make it worse.
   Therefore, Selective repeat window should be adopted as it sends the specific packet that may have been lost.

3. **Discarding Policy:**
   A good discarding policy adopted by the routers is that the routers may prevent congestion and at the same time partially discard the corrupted or less sensitive packages and also be able to maintain the quality of a message.
   In case of audio file transmission, routers can discard less sensitive packets to prevent congestion and also maintain the quality of the audio file.

4. **Acknowledgment Policy:**
   Since acknowledgements are also the part of the load in the network, the acknowledgment policy imposed by the receiver may also affect congestion. Several

approaches can be used to prevent congestion related to acknowledgment.
The receiver should send acknowledgement for N packets rather than sending acknowledgement for a single packet. The receiver should send an acknowledgment only if it has to send a packet or a timer expires.

5. **Admission Policy:**
   In admission policy a mechanism should be used to prevent congestion. Switches in a flow should first check the resource requirement of a network flow before transmitting it further. If there is a chance of a congestion or there is a congestion in the network, router should deny establishing a virtual network connection to prevent further congestion.

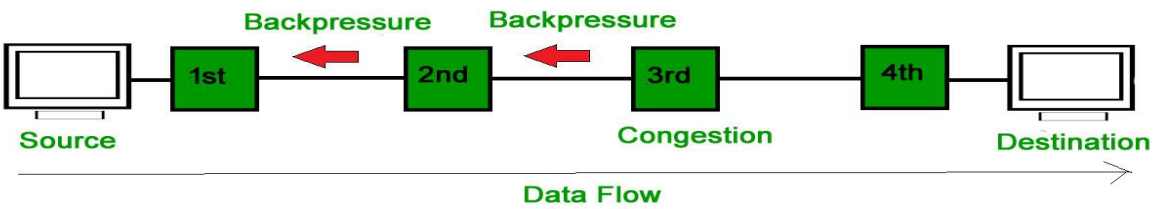
All the above policies are adopted to prevent congestion before it happens in the network.

Closed Loop Congestion Control

Closed loop congestion control techniques are used to treat or alleviate congestion after it happens. Several techniques are used by different protocols; some of them are:

1. Backpressure:

Backpressure is a technique in which a congested node stops receiving packets from upstream node. This may cause the upstream node or nodes to become congested and reject receiving data from above nodes. Backpressure is a node-to-node congestion control technique that propagate in the opposite direction of data flow. The backpressure technique can be applied only to virtual circuit where each node has information of its above upstream node.

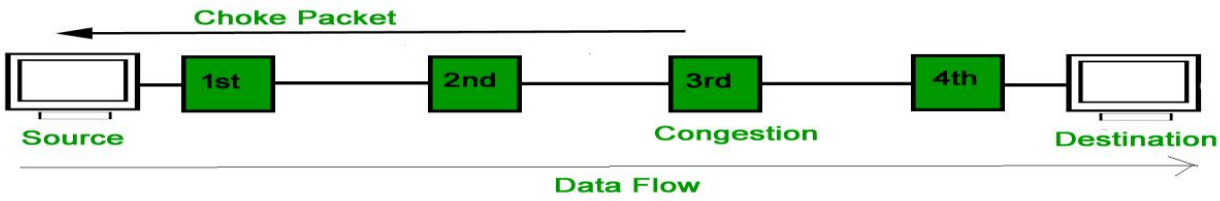


In above diagram the 3rd node is congested and stops receiving packets as a result 2nd node may be get congested due to slowing down of the output data flow. Similarly 1st

node may get congested and inform the source to slow down.

2. Choke Packet Technique:

Choke packet technique is applicable to both virtual networks as well as datagram subnets. A choke packet is a packet sent by a node to the source to inform it of congestion. Each router monitors its resources and the utilization at each of its output lines. Whenever the resource utilization exceeds the threshold value which is set by the administrator, the router directly sends a choke packet to the source giving it a feedback to reduce the traffic. The intermediate nodes through which the packets has traveled are not warned about congestion.



3. Implicit Signaling :

In implicit signaling, there is no communication between the congested nodes and the source. The source guesses that there is congestion in a network. For example when sender sends several packets and there is no acknowledgment for a while, one assumption is that there is a congestion.

4. Explicit Signaling :

In explicit signaling, if a node experiences congestion it can explicitly sends a packet to the source or destination to inform about congestion. The difference between choke packet and explicit signaling is that the signal is included in the packets that carry data rather than creating a different packet as in case of choke packet technique.

Explicit signaling can occur in either forward or backward direction.

- **Forward Signaling :** In forward signaling, a signal is sent in the direction of the congestion. The destination is warned about congestion. The receiver in this case adopt policies to prevent further congestion.

- **Backward Signaling :** In backward signaling, a signal is sent in the opposite direction of the congestion. The source is warned about congestion and it needs to slow down.

Network Address Translation (NAT)

- To access the Internet, one public IP address is needed, but we can use a private IP address in our private network.
- **The idea of NAT is to allow multiple devices to access the Internet through a single public address**.
- To achieve this, the translation of a private IP address to a public IP address is required.
- **Network Address Translation (NAT)** is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts.
- It also, does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination.
- It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.

Network Address Translation (NAT) working –

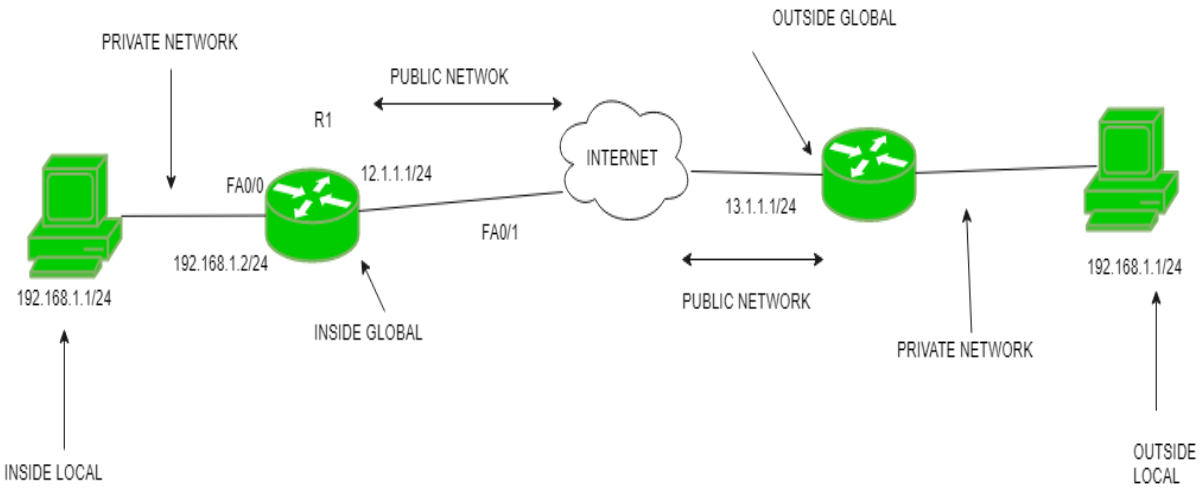
- Generally, the border router is configured for NAT i.e the router which has one interface in the local (inside) network and one interface in the global (outside) network.
- When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address.
- When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.
- Now If, NAT runs out of addresses, i.e., no address is left in the pool configured then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent.

Why mask port numbers?

- Suppose, in a network, two hosts A and B are connected. Now, both of them request for the same destination, on the same port number, say 1000, on the host side, at the same time.
- If NAT does only translation of IP addresses, then when their packets will arrive at the NAT, both of their IP addresses would be masked by the public IP address of the network and sent to the destination.
- Destination will send replies to the public IP address of the router.
- Thus, on receiving a reply, it will be unclear to NAT as to which reply belongs to which host (because source port numbers for both A and B are the same).
- Hence, to avoid such a problem, NAT masks the source port number as well and makes an entry in the NAT table.

NAT inside and outside addresses –

- **Inside refers to the addresses which must be translated.**
- **Outside refers to the addresses which are not in control of an organization**.
- These are the network Addresses in which the translation of the addresses will be done.



- **Inside local address –** An IP address that is assigned to a host on the Inside (local) network. The address is probably not an IP address assigned by the service provider i.e., these are private IP addresses. This is the inside host seen from the inside network.

- **Inside global address –** IP address that represents one or more inside local IP addresses to the outside world. This is the inside host as seen from the outside network.

- **Outside local address –** This is the actual IP address of the destination host in the local network after translation.

- **Outside global address –** This is the outside host as seen from the outside network. It is the IP address of the outside destination host before translation.

Network Address Translation (NAT) Types –

There are 3 ways to configure NAT:

1. **Static NAT –** In this, a single unregistered (Private) IP address is mapped with a legally registered (Public) IP address i.e one-to-one mapping between local and global addresses. This is generally used for Web hosting. These are not used in organizations as there are many devices that will need Internet access and to provide Internet access, a public IP address is needed.
   Suppose, if there are 3000 devices that need access to the Internet, the organization has to buy 3000 public addresses that will be very costly.

2. **Dynamic NAT –** In this type of NAT, an unregistered IP address is translated into a registered (Public) IP address from a pool of public IP addresses. If the IP address of the pool is not free, then the packet will be dropped as only a fixed number of private IP addresses can be translated to public addresses.
   Suppose, if there is a pool of 2 public IP addresses then only 2 private IP addresses can be translated at a given time. If 3rd private IP address wants to access the Internet then the packet will be dropped therefore many private IP addresses are mapped to a pool of public IP addresses. NAT is used when the number of users who want to access the Internet is fixed. This is also very costly as the organization has to buy many global IP addresses to make a pool.

3. **Port Address Translation (PAT) –** This is also known as NAT overload. In this, many local (private) IP addresses can be translated to a single registered IP address. Port

numbers are used to distinguish the traffic i.e., which traffic belongs to which IP address. This is most frequently used as it is cost-effective as thousands of users can be connected to the Internet by using only one real global (public) IP address.

Advantages of NAT –

- NAT conserves legally registered IP addresses.
- It provides privacy as the device's IP address, sending and receiving the traffic, will be hidden.
- Eliminates address renumbering when a network evolves.

Disadvantage of NAT –

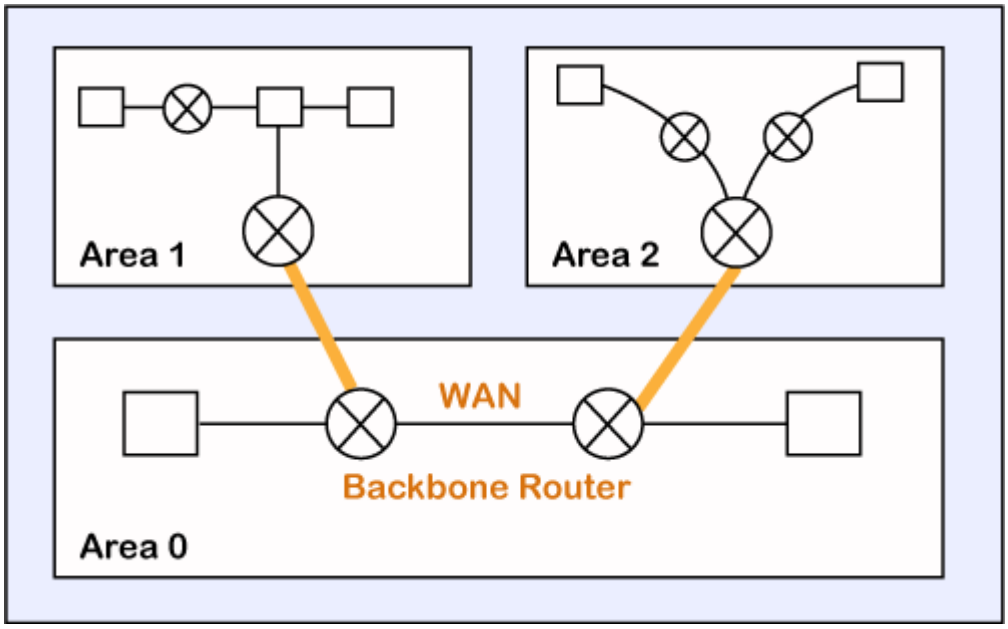
- Translation results in switching path delays.
- Certain applications will not function while NAT is enabled.
- Complicates tunneling protocols such as IPsec.
- Also, the router being a network layer device, should not tamper with port numbers(transport layer) but it has to do so because of NAT.

OSPF Protocol

- The OSPF stands for **Open Shortest Path First**.
- It is an intradomain protocol, which means that it is used within an area or a network.
- It is an interior gateway protocol that has been designed within a single autonomous system.
- It is based on a link-state routing algorithm in which each router contains the information of every domain, and based on this information, it determines the shortest path.

- The OSPF achieves by learning about every router and subnet within the entire network. Every router contains the same information about the network.
- The way the router learns this information by sending LSA (Link State Advertisements).
- These LSAs contain information about every router, subnet, and other networking information. Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB.
- The main goal is to have the same information about every router in an LSDBs.

OSPF Areas



- OSPF divides the autonomous systems into areas where the area is a collection of networks, hosts, and routers.
- Like internet service providers divide the internet into a different autonomous system for easy management and OSPF further divides the autonomous systems into Areas.
- Routers that exist inside the area flood the area with routing information
- In Area, the special router also exists. The special routers are those that are present at the border of an area, and these special routers are known as Area Border Routers.
- This router summarizes the information about an area and shares the information with other areas.
- All the areas inside an autonomous system are connected to the backbone routers, and these backbone routers are part of a primary area.

- The role of a primary area is to provide communication between different areas.

How does OSPF work?

There are three steps that can explain the working of OSPF:

Step 1: The first step is to become OSPF neighbors. The two connecting routers running OSPF on the same link creates a neighbor relationship.

Step 2: The second step is to exchange database information. After becoming the neighbors, the two routers exchange the LSDB information with each other.

Step 3: The third step is to choose the best route. Once the LSDB information has been exchanged with each other, the router chooses the best route to be added to a routing table based on the calculation of SPF.

How a router forms a neighbor relationship?

The first thing is happened before the relationship is formed is that each router chooses the router ID.

Router ID (RID): The router ID is a number that uniquely identifies each router on a network. The router ID is in the format of the IPv4 address. There are few ways to set the router ID, the first way is to set the router ID manually and the other way is to let the router decides itself.

The following is the logic that the router chooses to set the router ID:

- Manually assigned: The router checks whether the router ID is manually set or not. If it manually set, then it is a router ID. If it is not manually set, then it will choose the highest 'up' status loopback interface IP address. If there are no loopback interfaces, then it will choose the highest 'up' status non-loopback interface IP address.

Two routers connected to each other through point to point or multiple routers are connected can communicate with each other through an OSPF protocol. The two routers are adjacent only when both the routers send the HELLO packet to each other. When both the routers receive the acknowledgment of the HELLO packet, then they come in a two-way state. As OSPF is a link state routing protocol, so it allows to create the neighbor relationship between the routers. The

two routers can be neighbors only when they belong to the same subnet, share the same area id, subnet mask, timers, and authentication. The OSPF relationship is a relationship formed between the routers so that they can know each other. The two routers can be neighbors if atleast one of them is designated router or backup designated router in a network, or connected through a point-to-point link.

Types of links in OSPF

A link is basically a connection, so the connection between two routers is known as a link.

There are four types of links in OSPF:

1. **Point-to-point link:** The point-to-point link directly connects the two routers without any host or router in between.
2. **Transient link:** When several routers are attached in a network, they are known as a transient link.
   The transient link has two different implementations:
   Unrealistic topology: When all the routers are connected to each other, it is known as an unrealistic topology.
   Realistic topology: When some designated router exists in a network then it is known as a realistic topology. Here designated router is a router to which all the routers are connected. All the packets sent by the routers will be passed through the designated router.
3. **Stub link:** It is a network that is connected to the single router. Data enters to the network through the single router and leaves the network through the same router.
4. **Virtual link:** If the link between the two routers is broken, the administration creates the virtual path between the routers, and that path could be a long one also.

OSPF Packets

There are five different types of packets in OSPF:

- Hello
- Database Description
- Link state request

- Link state update
- Link state Acknowledgment

Let's discuss each packet in detail.

1. Hello packet

The Hello packet is used to create a neighborhood relationship and check the neighbor's reachability. Therefore, the Hello packet is used when the connection between the routers need to be established.

2. Database Description

After establishing a connection, if the neighbor router is communicating with the system first time, it sends the database information about the network topology to the system so that the system can update or modify accordingly.

3. Link state request

The link-state request is sent by the router to obtain the information of a specified route. Suppose there are two routers, i.e., router 1 and router 2, and router 1 wants to know the information about the router 2, so router 1 sends the link state request to the router 2. When router 2 receives the link state request, then it sends the link-state information to router 1.

4. Link state update

The link-state update is used by the router to advertise the state of its links. If any router wants to broadcast the state of its links, it uses the link-state update.

5. Link state acknowledgment

The link-state acknowledgment makes the routing more reliable by forcing each router to send the acknowledgment on each link state update. For example, router A sends the link state update to the router B and router C, then in return, the router B and C sends the link- state acknowledgment to the router A, so that the router A gets to know that both the routers have received the link-state update.

OSPF States

The device running the OSPF protocol undergoes the following states:

- **Down:** If the device is in a down state, it has not received the HELLO packet. Here, down does not mean that the device is physically down; it means that the OSPF process has not been started yet.
- **Init:** If the device comes in an init state, it means that the device has received the HELLO packet from the other router.
- **2WAY:** If the device is in a 2WAY state, which means that both the routers have received the HELLO packet from the other router, and the connection gets established between the routers.
- **Exstart:** Once the exchange between the routers get started, both the routers move to the Exstart state. In this state, master and slave are selected based on the router's id. The master controls the sequence of numbers, and starts the exchange process.
- **Exchange:** In the exchange state, both the routers send a list of LSAs to each other that contain a database description.
- **Loading:** On the loading state, the LSR, LSU, and LSA are exchanged.
- **Full:** Once the exchange of the LSAs is completed, the routers move to the full state.

Interior Gateway Routing Protocol (IGRP)

- Interior Gateway Routing Protocol (IGRP) is a proprietary distance vector routing protocol used to communicate routing information within a host network.
- It was invented by Cisco.
- IGRP manages the flow of routing information within connected routers in the host network or autonomous system.
- The protocol ensures that every router has routing tables updated with the best available path.
- IGRP also avoids routing loops by updating itself with the changes occurring over the network and by error management.
- Cisco created Interior Gateway Routing Protocol (IGRP) in response to the limitations in Routing Information Protocol (RIP), which handles a maximum hop count of 15.

- IGRP supports a maximum hop count of up to 255. The primary two purposes of IGRP are to:
    - **Communicate routing information to all connected routers within its boundary or autonomous system**
    - **Continue updating whenever there is a topological, network or path change that occurs**
- IGRP sends a notification of any new changes, and information about its status, to its neighbors every 90 seconds.
- IGRP manages a routing table with the most optimal path to respective nodes and to networks within the parent network.
- Because it is a distance vector protocol, IGRP uses several parameters to calculate the metric for the best path to a specific destination.
- These parameters include delay, bandwidth, reliability, load and maximum transmission unit (MTU).

Characteristics

The following are the characteristics of the IGRP (Interior Gateway Routing Protocol):

1. The Internet Group Routing Technology (IGRP) is a distance-vector routing protocol created by Cisco.
2. In addition to bandwidth, delay (by default), reliability, load, and MTU are all measured in the IGRP protocol.
3. It transmits updates every 90 seconds, with a hold-down time of 280 seconds between each broadcasting session.
4. When network changes occur, triggered updates are utilized to expedite the convergence process.
5. The IGRP router command needs the inclusion of an AS number.
6. For routers to communicate routing information, they must be in the same Associated System Number (AS).
7. The maximum number of hops allowed by IGRP is 255. It has a default value of 100 and is often changed to 50 or less.
8. The IGRP AD value is 100.

Advantages:

1. The procedure is simple and uncomplicated.
2. It takes into account the latency, bandwidth, reliability, and load of a network connection while calculating the score. Consequently, it is quite accurate when it comes to selecting the most suited approach.
3. The use of composite metrics
4. Configuration is straightforward.
5. When compared to RIP, it has more scalability (255 hops, 100 by default)

Disadvantages:

1. The hop count is limited to 15; if a packet has traveled through 15 routers and still has another router to travel to, it will be discarded.
2. Does not support a variable-length subnet mask (VLSM), which means that it sends routing updates based only on a fixed-length subnet mask (FLSM) or routes that fall on classful boundaries. As a result, RIP V1 will not function on a network that has been subnetted beyond the standard /8, /16, and /24 (255.0.0.0, 255.255.255.0) or Class A, B, and C network borders (255.0.0.0, 255.255.255.0).
3. Convergence occurs slowly, particularly on large networks.
4. Doesn’t know how much bandwidth is available on a given connection.
5. Doesn’t allow numerous pathways for the same route
6. Routing updates may use a substantial amount of bandwidth since the whole routing database is delivered whenever the state of a connection changes. Routers are prone to routing loops.

Enhanced Interior Gateway Routing Protocol (EIGRP)

- It is a dynamic routing protocol that is used to find the best path between any two-layer 3 devices to deliver the packet.
- EIGRP works on network layer Protocol of OSI model and uses protocol number 88. It uses metrics to find out the best path between two layer 3 devices (router or layer 3 switches) operating EIGRP.

- Administrative Distance for EIGRP are:-

EIGRP routes	AD values
Summary Routes	5
Internal Routes	90
external routes	170

- Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco-proprietary Hybrid routing protocol that contains features of distance-vector and link-state routing protocols.
- It is a network layer protocol that works on protocol number 88.

Some of its features are:

1. **Rapid convergence –** EIGRP uses a DUAL algorithm to support rapid convergence. If a route to a network goes down then another route(feasible successor) can be used. If there is no route present to that network in the topology table also then a query message is multicast to find out the alternative route to that network.

2. **Reduced bandwidth usage –** EIGRP doesn't send periodic updates like other distance vector routing protocol does. Distance Vector Routing protocol like RIP sends full routing table over a period of time, therefore, consumes the available bandwidth needlessly but EIGRP uses partial updates if there is any change in the topology occurs i.e updates are triggered only if any event occurs therefore consuming the bandwidth when needed. Also, EIGRP updates are propagated to the routers only who require it.

3. **Support all LAN and WAN data link protocols and topologies –** EIGRP supports multi-access networks like FDDI, token ring, etc, and all WAN topologies like leased line, point-to-point links. EIGRP doesn't require any

additional configuration across layer 2 protocols like frame relay.

4. **Supports auto-summary –** In EIGRP, auto-summarization is enabled by default. Auto summarization is a feature that allows Routing Protocols to summarize their routes to their classful networks automatically i.e routers will receive summarised routes automatically. EIGRP. e.g-1.1.1.1 /24 will be automatically summarised to the classful 1.1.1.1/8

5. **Supports unequal cost load balancing –** Unequal cost load balancing is possible in EIGRP by changing the value of variance. By default, variance is 1, therefore, supports equal-cost load balancing but if we want to use unequal cost load balancing then we can change the value of variance according to the amount of traffic we want to divide across different paths. Feasible distance is multiplied in such a way that it becomes greater than the value of the feasible distance of successor.

6. **Communication via Reliable Transfer Protocol (RTP) –** EIGRP depends upon proprietary protocol RTP to manage the communication between EIGRP speaking routers. EIGRP uses 224.0.0.10 as its a multicast address. For each multicast it sends, the router prepares and maintains a list of routers (speaking EIGRP). If no acknowledgement of multicast is received then the same data is transmitted through 16 unicast messages. If no acknowledgement is received even after 16 unicast attempts then it is declared dead. This process is known as reliable multicast.

7. **Best path selection using DUAL –** EIGRP uses Diffusing Update Algorithm (DUAL) to find out the best path available to a network. EIGRP speaking routers maintain a topology table in which all the routes to the network are maintained. If the best path (successor) goes down, then the second best path (feasible successor) is used from the topology table. If there is no path available in the topology table then it sends a query message to resolve the query.

It maintains 3 different tables mainly:

(a) Neighbor table: It contains information about the routers with which neighbourship has been formed. It contains the SRTT, RTP. It also contains the queue count value for the hello messages that are not being acknowledged.
(b) Topology table: It contains all the routes available to a network (both feasible successor and successor).
(c) Routing table: It contains all the routes which are being used to make current routing decisions. The routes in this table are considered as successor (best path) routes.

8. **Traffic control –** Suppose if one of the interfaces of the router is connected to ISP then we don't want that interface to be part of the EIGRP process. For this scenario, EIGRP provides a feature in which we can flag the interface as passive i.e not to take part in the EIGRP process.

9. Support Variable Length Subnet Mask (VLSM).
10. Support for both IPv4 and IPv6.

- It uses some messages to communicate with the neighbour devices that operate EIGRP. These are:-

1. **Hello message-**These messages are kept alive messages which are exchanged between two devices operating EIGRP. These messages are used for neighbour discovery/recovery, if there is any device operating EIGRP or if any device(operating EIGRP) coming up again.
   These messages are used for neighbor discovery if multicast at 224.0.0.10. It contains values like AS number, k values, etc.
   These messages are used as acknowledgement when unicast. A hello with no data is used as the acknowledgement.

2. **NULL update-**It is used to calculate SRTT(Smooth Round Trip Timer) and RTO(Retransmission Time Out).

   SRTT: The time is taken by a packet to reach the neighboring router and the acknowledgement of the packet to reach the local router.

   RTO: If a multicast fails then unicast is being sent to that router. RTO is the time for which the local router waits for an acknowledgement of the packet.
3. **Full Update –** After exchanging hello messages or after the neighbourship is formed, these messages are exchanged. This message contains all the best routes.
4. **Partial update-**These messages are exchanged when there is a topology change and new links are added. It contains only the new routes, not all the routes. These messages are multicast.
5. **Query message-**These messages are multicast when the device is declared dead and it has no routes to it in its topology table.
6. **Reply message –** These messages are the acknowledgment of the query message sent to the originator of the query message stating the route to the network which has been asked in the query message.
7. **Acknowledgement message**

   It is used to acknowledge EIGRP updates, queries, and replies. Acks are hello packets that contain no data.

   Note:-Hello and acknowledgment packets do not require any acknowledgment.

   Reply, query, update messages are reliable messages i.e require acknowledgement.

Composite matrix-The EIGRP composite metric calculation can use up to 5 variables, but only 2 are used by default (K1 and K3). The composite metric values are :

K 1 (bandwidth)

K 2 (load)

K 3 (delay)

K 4 (reliability)

K 5 (MTU)

The lowest bandwidth, load, delay, reliability, MTU along the path between the source and the destination is considered in the composite matrix in order to calculate the cost.

Note:- Generally, only k1 and k3 values are used for metric calculation by EIGRP. The values are 10100 for k1, k2, k3, k4, k5 respectively.

criteria To form EIGRP neighbourship, these criteria should be fulfilled:-

1. k values should match.
2. Autonomous system number should match. (AS is a group of networks running under a single administrative control) .
3. authentication should match (if applied). EIGRP supports MD5 authentication only.
4. subnet mask should be the same.

Timers:-

Hello timer- The interval in which EIGRP sends a hello message on an interface. It is 5 seconds by default.

Dead timer- The interval in which the neighbor will be declared dead if it is not able to send the hello packet. It is 15 seconds by default.

Border Gateway Protocol (BGP)

- Border Gateway Protocol (BGP) is used to Exchange routing information for the internet and is the protocol used between ISP which are different ASes.
- The protocol can connect together any internetwork of autonomous system using an arbitrary topology.
- The only requirement is that each AS have at least one router that is able to run BGP and that is router connect to at least one other AS's BGP router. BGP's main function is to exchange network reach-ability information with other BGP systems.
- Border Gateway Protocol constructs an autonomous systems' graph based on the information exchanged between BGP routers.

Characteristics of Border Gateway Protocol (BGP):

- **Inter-Autonomous System Configuration:** The main role of BGP is to provide communication between two autonomous systems.
- BGP supports Next-Hop Paradigm.
- Coordination among multiple BGP speakers within the AS (Autonomous System).
- **Path Information:** BGP advertisement also include path information, along with the reachable destination and next destination pair.

- **Policy Support:** BGP can implement policies that can be configured by the administrator. For ex:- a router running BGP can be configured to distinguish between the routes that are known within the AS and that which are known from outside the AS.
- Runs Over TCP.
- BGP conserve network Bandwidth.
- BGP supports CIDR.
- BGP also supports Security.

Functionality of Border Gateway Protocol (BGP):

BGP peers performs 3 functions, which are given below.

1. The first function consist of initial peer acquisition and authentication. both the peers established a TCP connection and perform message exchange that guarantees both sides have agreed to communicate.
2. The second function mainly focus on sending negative or positive reach-ability information.
3. The third function verifies that the peers and the network connection between them are functioning correctly.

BGP Route Information Management Functions:

- **Route Storage:** Each BGP stores information about how to reach other networks.
- **Route Update:** In this task, Special techniques are used to determine when and how to use the information received from peers to properly update the routes.
- **Route Selection:** Each BGP uses the information in its route databases to select good routes to each network on the internet network.
- **Route advertisement:** Each BGP speaker regularly tells its peer what is knows about various networks and methods to reach them.

BASIS FOR COMPARISON	EIGRP	OSPF
Stands for	Enhanced Interior Gateway Protocol	Open Shortest Path First
Standards-based on	Cisco Proprietary	IETF Open Standard
Protocol type	Hybrid	Link state
Routing metrics	Combination of bandwidth, reliability, load and delay.	Interface bandwidth
Administrative distance	90 (Internal) 170 (External)	110
CPU requirements	Lower CPU and memory needs	Require high CPU and memory
Algorithm	DUAL distance vector	Dijkstra link state

BASIS FOR COMPARISON	EIGRP	OSPF
Hierarchical design	No	Yes
Support of IPX and AppleTalk	Yes	No
Updates	Updates and queries as needed to a multicast address	Flooding as needed and periodically to a multicast address
Ease of implementation	Easy but no provision of auto-summary	Complicated
Loop prevention	Split horizon and DUAL	Full knowledge of topology
Filtering and summarization	Possible anywhere in the network	Only on ASBRs or ABRs